

Doppler flow velocity waveform analysis in high risk pregnancies: a randomized controlled trial

JOHN P. NEWNHAM, MICHELLE R-A. O'DEA, KAREN P. REID,
DEAN A. DIEPEVEEN

Abstract

Objective—To test whether the introduction of Doppler waveform analysis into the ultrasound department of a tertiary level hospital reduces neonatal morbidity and improves obstetric management.

Design—A randomized controlled trial.

Setting—Department of Ultrasound, King Edward Memorial Hospital, Perth, Western Australia.

Subjects—505 women with pregnancy abnormalities referred to an ultrasound department for fetal investigation during the third trimester.

Intervention—Continuous wave Doppler studies of umbilical and utero-placental arterial circulations. Results were revealed to patients and clinicians.

Main outcome measures—Principal end point was the duration of neonatal stay in hospital; other end points included the number and type of fetal heart rate monitoring studies, obstetric interventions, frequency of fetal distress, birthweight, Apgar scores and need for neonatal intensive care.

Results—There was no effect on the duration of neonatal stay in hospital. Small trends in obstetric management were observed with study group patients having fewer contraction stress tests, less likelihood of antepartum fetal distress, and more likelihood of fetal distress after induction of labour leading to emergency caesarean section. Depressed Apgar scores were more frequent in the study group.

Conclusion—Introduction of Doppler waveform studies did not result in reduced neonatal morbidity but did have a small effect on obstetric management. For each institution the role of Doppler studies in late pregnancy will be influenced by the usage of other tests of fetal welfare already entrenched in clinical practice.

In recent years examination of umbilical and uteroplacental arterial waveforms by Doppler

University Department of Obstetrics and
Gynaecology, King Edward Memorial Hospital for
Women, Perth, Western Australia

J. P. NEWNHAM Associate Professor

M. R-A. O'DEA Research Midwife

K. P. REID Research Midwife

D. A. DIEPEVEEN Research Officer

Correspondence: John P. Newnham MD, King
Edward Memorial Hospital for Women, 374 Bagot
Road, Subiaco, Perth, Western Australia 6008.

ultrasound has presented a new and exciting method of investigation in the management of pregnancy. Continuous wave Doppler systems provide an indirect estimate of placental vascular resistance (Giles *et al.* 1985). They are relatively inexpensive and technically easy to use. It is tempting to speculate that introduction of this technology into clinical practice will improve clinical management and reduce perinatal morbidity. However, the availability of new technology does not necessarily justify its use. Justification for widespread acceptance of any new test requires demonstration of its efficacy in

prospective studies, followed by evidence from randomized controlled trials of improved clinical outcome (Nelson 1987). When continuous wave Doppler systems first became available for purchase in 1985, the obstetric and ultrasound communities in Western Australia agreed to withhold this technology in the state until its value had been demonstrated in rigorously designed studies conducted locally. The initial investigation was a prospective double blind study of more than 500 pregnancies which determined the predictive power of Doppler waveform systolic/diastolic ratios at a variety of gestational ages, and related these findings to the results of conventional ultrasound biometry (Newnham *et al.* 1990a). The second study, reported here, was a randomized controlled trial designed to assess the effects of introducing a continuous wave Doppler system into the ultrasound department of a tertiary level hospital. Until the commencement of this trial, results of obstetric Doppler studies had not been revealed to clinicians or patients in Western Australia. The principal purpose of this study was to assess the role of Doppler studies during the third trimester in pregnancies in which the fetus had

already been shown to be at risk, since our previous study had shown the value of this test in the prediction, during the third trimester, of the fetus destined to suffer the combination of growth retardation and fetal distress.

Subjects and methods

Enrolment

Pregnant women referred to the ultrasound department of King Edward Memorial Hospital for Women, Perth, Western Australia between January 1989 and April 1990 were invited to participate in the study. This hospital is the sole tertiary perinatal centre for the state of Western Australia. Criteria for enrolment in the study included those disorders of pregnancy in which an increased risk of retarded fetal growth or impaired fetal wellbeing were considered likely (Table 1). The study had been approved by the hospital's ethics committee. Eligibility for enrolment was assessed on completion of each initial ultrasound examination and written consent was then obtained from each participant.

Table 1. Details of the women in the study and control groups

	Study (<i>n</i> = 254)	Control (<i>n</i> = 251)
Age (years)	26.6 (7.3)	26.4 (5.6)
Height (cm)	163 (6.7)	162 (7.3)
Parity 0	112 (44.1)	97 (38.6)
1	72 (28.3)	72 (28.7)
2	31 (12.2)	41 (16.3)
3 or more	39 (15.3)	41 (16.4)
Smoking status		
Non-smoker	154 (60.6)	175 (69.7)
<20/day	76 (29.9)	51 (20.3)
≥20/day	24 (9.4)	25 (10.0)
Indications for recruitment:		
Pre-eclampsia	30 (14.9)	51 (20.3)
Small for gestational age	43 (16.9)	45 (17.9)
Preterm labour	27 (10.6)	29 (11.6)
Diabetes	20 (7.9)	25 (10.0)
Antepartum haemorrhage	39 (15.4)	23 (9.2)
Twin gestation	21 (8.3)	19 (7.6)
Prelabour rupture of membranes	28 (11.0)	16 (6.4)
Other*	38 (15.0)	43 (17.1)
Gestational age at enrolment (weeks)	32.7 (4.0)	32.6 (3.9)

Results are *n* (%) or mean (SD) as appropriate.

* Other includes poor obstetric history, oligohydramnios, polyhydramnios and chronic hypertension.

Randomization

Women were then randomized by draw of instructions from numbered, opaque, sealed envelopes either to have ($n = 254$) or to not have ($n = 251$) Doppler waveform studies. Power calculation estimated that a total sample of 500 women would have an 80% power of detecting a 0.5 day reduction in median neonatal hospital stay ($\alpha = 0.05$; two tailed). Randomization was stratified for twin gestations. In those pregnancies randomized to receive waveform studies the first study was performed immediately and the frequency of subsequent examinations was determined by the ultrasonologist and the referring clinician.

Doppler examinations

Doppler flow velocity waveform studies were performed with a continuous wave system (Medasonics SP25A, Mountain View, CA and a D10 bidirectional Doppler) as described previously (Newnham *et al.* 1990a). The subjects were supine with lateral tilt provided by a wedge under one hip. The ratio of peak systolic (S) to least diastolic (D) Doppler shift frequency was calculated from waveforms obtained from an umbilical artery and from a maternal uteroplacental artery within the placental bed. These ratios were not adjusted to standard fetal or maternal heart rates (Newnham *et al.* 1990b). Results of the waveform studies were placed in the hospital records and reported both as numerical values and graphically on reference ranges prepared with data published elsewhere (Thompson *et al.* 1988). The clinician in charge of each case was also informed by telephone of any result which was outside the reference range.

Clinical management

Before the start of the study an educational programme was provided for clinicians whose patients might be eligible for enrolment. This programme included a written summary of the literature on Doppler flow studies, formal lectures and frequent hospital meetings and tutorials. Low dose aspirin therapy was not prescribed for any of the patients in this study. Antenatal fetal heart rate (FHR) monitoring recordings were classified using criteria reported previously (Newnham *et al.* 1988).

Statistics

The diagnosis of hypoxic ischaemic encephalopathy

was made retrospectively by review of hospital records by a paediatrician not otherwise involved in the study and who was unaware of the results of randomization of the woman and her baby (Sarnat & Sarnat 1976).

Comparison of means between groups was by Student's *t*-test. Logarithmic transformation was employed for data with skewed distributions. Where appropriate, outcome variables are presented as re-transformed means with 95% confidence intervals. The 95% CI of the odds ratios were calculated using the method described by Katz *et al.* (1978). An interim analysis was not performed.

Results

Characteristics of the women and their pregnancies are shown in Table 1. There were 21 twin pregnancies in the study group and 19 in the control group resulting in a total of 275 and 270 fetuses in the study and control groups respectively. Analysis after randomization showed that the two groups were comparable for maternal age, height, parity, smoking and gestational age at enrolment. The most frequent indications for enrolment were pre-eclampsia, ultrasonic evidence of fetal growth retardation, antepartum haemorrhage and preterm labour.

Details of antenatal tests of fetal wellbeing are shown in Table 2. In eight pregnancies the last umbilical artery waveform study indicated absent end-diastolic flow and in a further 57 the last umbilical artery S-D ratio was >95th centile although diastolic flow was present. Of the eight fetuses with absent end-diastolic flow, one died *in utero* at 29 weeks gestation (birthweight 870 g) the day before the attending clinician had scheduled FHR monitoring to commence, two infants born at 26 weeks gestation (birthweights 465 and 555 g) died in the neonatal period and five survived with stays in neonatal intensive care ranging from 27 to 89 days (28–35 weeks gestation, 770–1320 g). In each of these eight pregnancies the fetal abdominal circumference measurement was <5th centile and five of them were associated with oligohydramnios. All eight pregnancies were known to have fetal growth retardation and six had features of pre-eclampsia. Seven pregnancies were delivered by caesarean section, six for fetal distress, diagnosed on FHR monitoring, and one for maternal pre-eclampsia.

Of the 57 fetuses with elevated umbilical

Table 2. Details of antenatal tests of fetal wellbeing in the study and control groups

	Study (<i>n</i> = 254)	Control (<i>n</i> = 251)	Odds ratio (95% CI)
Number of Doppler studies			
0	0	251 (100)	
1	152 (59.8)	0	
2	60 (23.6)	0	
3 or more	42 (16.5)	0	
Umbilical artery S/D ratio >95th centile	57 (22.4)	—	
Absent umbilical artery diastolic flow	8 (3.2)	—	
Uteroplacental S/D ratio >95th centile	9 (3.5)	—	
Fetal abdominal circumference <5th centile	56 (22.0)	44 (17.5)	1.33 (0.86–2.07)
Antenatal FHR monitoring studies			
0	40 (15.7)	43 (17.1)	0.90 (0.56–1.45)
1–2	82 (32.2)	73 (29.0)	1.16 (0.80–1.70)
3–5	66 (26.0)	78 (31.1)	0.78 (0.53–1.15)
more than 5	66 (26.0)	57 (22.8)	1.19 (0.80–1.80)
Last FHR monitoring test			
Non-stress test	209 (97.6)	196 (94.2)	1.30 (0.84–2.02)
Contraction stress test	5 (2.4)	12 (5.8)	0.40 (0.14–1.15)

Results are *n* (%).

artery S-D ratios, but with diastolic flow, 35 (61%) had been shown previously to have reduced abdominal circumference measurements. Thirty-six (63%) had birthweights <10th centile. Labour was induced in 29 of the 57 pregnancies with elevated S-D ratios, of which eight (28%) developed fetal distress requiring emergency caesarean section. Eight further pregnancies were delivered by elective caesarean section.

Of the nine pregnancies with uteroplacental S-D ratios >95th centile, two were twin pregnancies in which there was a single placenta. Each of the four twin fetuses had a birthweight <10th centile, but had not shown features of fetal distress. Of the seven singleton fetuses with uteroplacental S-D ratios >95th centile, five had already been shown to have reduced abdominal circumference measurements and all but one had a birthweight <10th centile. The umbilical artery S-D ratios were abnormal in seven of these 11 fetuses.

The frequency of antenatal FHR monitoring studies was similar in the two groups. However, the last study was less likely to have been a con-

traction stress test in the study group than in the control group (OR 0.40, 95% CI 0.14 to 1.15). Of these contraction stress tests, one of five in the study group was abnormal, and five of 12 in the control group were abnormal.

The pregnancy outcomes are shown in Table 3. There were no statistically significant differences between the two groups in the duration of inpatient stay after recruitment or gestational age at delivery. A small trend towards more frequent spontaneous onset of labour was observed in the study group (OR 1.29, 95% CI 0.90 to 1.86) although this was not accompanied by differences between the two groups in the rates of induction of labour or of elective caesarean sections. Fetal distress after induction of labour (OR 1.38, 95% CI 0.81 to 2.37) leading to delivery by emergency caesarean section (OR 1.28, 95% CI 0.85 to 1.94) was more frequent in the study group, while the diagnosis of fetal distress before labour was less frequent (OR 0.76, 95% CI 0.41 to 1.43). However, these trends in obstetric management were small.

Details of the fetal outcomes are given in Table 4. Total neonatal days in hospital, birth-

Table 3. Comparison of the pregnancy outcomes in the study and control groups

	Study (<i>n</i> = 254)	Control (<i>n</i> = 251)	Odds ratio (95% CI)
Inpatient days after recruitment	4 (1-31) <i>P</i> = 0.857	4 (1-24)	
Gestational age at delivery (weeks)	36.9 (3.7) <i>P</i> = 0.471	37.1 (3.6)	
Onset of labour			
No labour	58 (22.8)	66 (26.3)	0.83 (0.55-1.24)
Spontaneous	100 (39.4)	84 (33.5)	1.29 (0.90-1.86)
Induction	96 (37.8)	101 (40.2)	0.90 (0.63-1.29)
Mode of delivery			
Vaginal	160 (63.0)	162 (64.5)	0.94 (0.65-1.34)
Elective caesarean section	29 (11.4)	36 (14.3)	0.77 (0.46-1.30)
Emergency caesarean section	65 (25.6)	53 (21.1)	1.28 (0.85-1.94)
Fetal distress			
Antepartum	19 (7.5)	24 (9.6)	0.76 (0.41-1.43)
After spontaneous onset of labour	15 (5.9)	13 (5.2)	1.15 (0.54-2.47)
After induction of labour	35 (13.8)	26 (10.4)	1.38 (0.81-2.37)

Data are *n* (%), mean (SD) or medians (95% CI) as appropriate.

weights and birthweight centiles were similar in the two groups. Apgar scores <7 were more frequent in the study group than in the control group both at 1 min (OR 1.62, 95% CI 1.10 to 2.38) and at 5 min (OR 2.55, 95% CI 0.97 to 6.67). In the study group, two infants developed hypoxic ischaemic encephalopathy and a third had an intraventricular haemorrhage. One infant in the control group developed periventricular leucomalacia. However, there were no differences between the two groups in any other measure of neonatal outcome, including requirement for ventilatory support or oxygen, the rate of admission to the neonatal intensive care unit, perinatal mortality or the frequency of congenital abnormality. In both the study and the control group, two of the nine perinatal deaths occurred in infants born with major congenital abnormalities. Review of the individual case records of those pregnancies in which low Apgar scores were recorded failed to reveal any consistent factor which could account for the greater frequency of low Apgar scores in the study group.

Discussion

The results of this study have confirmed that abnormal systolic/diastolic ratios of umbilical and uteroplacental arterial Doppler waveforms recorded during the third trimester are associ-

ated with poor pregnancy outcome. However, although certain effects on obstetric management were observed, introduction of these tests into clinical practice did not reduce neonatal morbidity. In the first randomized controlled trial of continuous wave Doppler studies, also performed in Australia, Trudinger *et al.* (1987) reported that the test provided clinicians with data which improved obstetrical decision making. Fetal distress and emergency caesarean section were less frequent in their study group, suggesting that Doppler results had identified those fetuses which would not tolerate labour; however, a corresponding increase in caesarean sections before the onset of labour in those who had received Doppler studies was not observed. A small reduction was seen in admission rates to neonatal intensive care, although the difference between the study and control groups was not statistically significant. The study by Trudinger *et al.* (1987) differed from the present study in that the results of Doppler tests had previously been available for clinical usage, which rendered their trial an investigation of the effects of withdrawing the test in the control group. In our study Doppler test results had not previously been released, making this an investigation of the effects of introducing the test.

In the present study, small but not statistically significant trends were observed in obstetric management which reflected differing appli-

Table 4. Comparison of the fetal outcomes in the study and control groups

	Study (n = 275)	Control (n = 270)	Odds ratio (95% CI)
Total neonatal days in hospital	6 (2-41) <i>P</i> = 0.78	6 (2-39)	
Birthweight (g)	2697 (860) <i>P</i> = 0.52	2745 (861)	
Birthweight centiles			
>90th	25 (9.1)	23 (8.5)	1.07 (0.59-1.94)
10-90th	157 (57.1)	158 (58.5)	0.94 (0.67-1.33)
<10th	93 (33.8)	89 (33.0)	1.04 (0.73-1.48)
Congenital abnormality	22 (8.0)	19 (7.0)	1.15 (0.61-2.17)
Perinatal death			
Stillbirth	3 (1.1)	2 (0.7)	1.48 (0.24-8.92)
Neonatal	6 (2.2)	7 (2.6)	0.84 (0.28-2.53)
Apgar score <7			
1 min	84 (31.6)	58 (22.2)	1.62 (1.10-2.38)
5 min	15 (3.3)	6 (2.6)	2.55 (0.97-6.67)
Required ventilatory support	37 (13.6)	29 (10.8)	1.29 (0.77-2.17)
Required oxygen	41 (15.1)	43 (16.0)	0.93 (0.58-1.47)
Admitted to neonatal intensive care	103 (37.5)	106 (39.3)	0.93 (0.66-1.31)

Data are *n* (%), mean (SD) or medians (95% CI) as appropriate.

cations of other tests of fetal welfare in the study and control groups. Pregnancies in which Doppler studies had been performed were less likely to have had a contraction stress test, had a lesser chance of receiving a prelabour diagnosis of fetal distress, and had a greater chance of fetal distress after induction of labour leading to emergency caesarean section. These findings support the clinical impression observed during the trial that the availability of Doppler studies altered the cascade of fetal investigations in general use in this institution. For women in the control group, the finding of a persistent non-reactive non-stress test was likely to lead to a contraction stress test, thus increasing the likelihood that a diagnosis of antepartum fetal distress would be made. For women in the study group with a persistent non-reactive non-stress test the availability of Doppler results may have diminished the likelihood that the clinician would seek extra information in the form of a contraction stress test, and if a diagnosis of fetal distress was then to be made, it would follow induction of labour. This usage of Doppler results contrasts with the effects on management observed by Trudinger *et al.* (1987) and probably reflects differences in protocols of fetal investigation. For each institu-

tion the effects of introducing Doppler flow studies on usage of other tests of fetal wellbeing will be influenced largely by local protocols and clinical practice.

Also, the liberal use of fetal abdominal circumference measurement in the present study may have diminished the opportunity for Doppler studies to improve pregnancy outcome. In our previous efficacy study (Newnham *et al.* 1990a), it was found that abdominal circumference and umbilical artery Doppler S-D ratios at 28 and 34 weeks gestation had impressive and comparable powers to predict the combination of fetal growth retardation and distress, despite the obvious differences in the nature of the two measurement systems. The design of the present study ensured that recruitment occurred after full ultrasound biometry had been performed, since it was intended that the study should address the role of this technology within an ultrasound department, rather than its role as a competitor with ultrasound measurement of fetal size. Because umbilical artery Doppler waveform studies and fetal abdominal circumference measurements both identify those fetuses destined to be growth retarded and distressed, it is vital that reports of trials in this field

describe the relative roles that each of these two tests has made on clinical management.

We are unable to offer an explanation for the greater frequency of low Apgar scores observed in the study group. This finding was not supported by other measured outcomes of neonatal welfare, could not be explained by the slightly greater proportion of growth retarded fetuses in the study group and had no consistent pattern when individual case records were examined. It is most unlikely that the scoring was biased by knowledge of the group to which the pregnancy had been allocated and we suggest that the differences in Apgar scores between the two groups may have been chance effects.

Previously reported management trials of Doppler studies have suggested that some neonatal outcome measures may be improved but interpretation is complicated by differing designs of the studies. McParland & Pearce (1988) described in a review article the results of a study of 509 pregnancies in which patients were stratified into 'concealed' or 'revealed' groups according to whether the waveforms were normal or abnormal. Fewer neonatal deaths were observed in the 'revealed' group although further details were not provided. In a randomized comparison of routine versus highly selective Doppler waveform and biophysical profile usage, Tyrell *et al.* (1990) observed fewer depressed 5 min Apgar scores and less neonatal morbidity in the routinely monitored group.

The results of the various trials of Doppler studies now present an incomplete answer to the question whether or not Doppler studies should be introduced into obstetric practice. There is little doubt that these tests provide information which is of use in obstetric management and the available data suggest that their introduction does not increase the number of preterm births or obstetric interventions. Ideally, justification for new forms of technology in this field requires demonstration that the test will reduce perinatal morbidity. Evidence of improved outcome from the use of Doppler waveform analysis in late pregnancy remains scanty. However, it may be unrealistic to expect that a new test of fetal welfare applied in late pregnancy would result in a dramatic improvement in outcome. In the management of fetuses identified as being growth retarded or at risk of hypoxia in late pregnancy, any contribution by yet another test will be lessened by the use of those tests already entrenched in clinical practice. For each institu-

tion, the extent of that contribution will be determined by the availability of other biophysical tests of fetal wellbeing and the competence with which they are applied. Moreover, evidence is emerging which suggests that many pregnancy disorders originate at conception or in early gestation and that perinatal events contribute less to long-term morbidity than was once thought. Cerebral palsy appears to be at least one example (Blair & Stanley 1988), although less severe neonatal disorders may also be included (Editorial 1989). Thus, major improvements in outcome may not necessarily result from further refinements in fetal assessment during late pregnancy. The identification in earlier pregnancy of those fetuses at risk of placental disease and growth retardation, leading to application of therapies aimed at improving growth and development (Trudinger *et al.* 1988; McParland *et al.* 1990), is a role in which Doppler technology is more likely to make a worthy and longstanding contribution.

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